

InfoHub — API Services, Subscriptions & Running Costs

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Executive Summary

InfoHub aggregates real-time data from **29 cryptocurrency exchanges** and multiple market data providers. The vast majority of exchange data comes from **free public APIs**. The primary paid services are hosting (Vercel), database (DigitalOcean PostgreSQL), and market data enrichment (CoinMarketCap).

Current plan: **Vercel Hobby (free)** — confirmed from deployment token.

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1. Infrastructure Services

1.1 Vercel — Hosting & Edge Runtime

Detail	Value
Current Plan	Hobby (Free)
Region	Dubai (dxb1)
Cron Jobs	4 (snapshot 6h, portfolio 12h, liquidations 1min, alerts 5min)
Functions	30s max duration
Free Includes	100GB bandwidth, 100K function invocations

Upgrade Path for 500 DAU:

Plan	Cost	Bandwidth	Functions	Notes
Hobby	\$0/mo	100 GB	100K invocations	Current — may hit limits
Pro	\$20/mo	1 TB + \$20 credit	10M edge requests	Recommended for 500 DAU
Pro (realistic)	\$20–80/mo	With overages	Serverless compute extra	Depends on API call volume

Estimate for 500 DAU: \$20–60/mo

1.2 DigitalOcean Managed PostgreSQL

Detail	Value
Current Tier	Basic (estimated \$15/mo single node)
Host	infohub-db-do-user-*.e.db.ondigitalocean.com

SSL	Required
Used For	Users, alerts, portfolio snapshots, liquidation history, funding history, OI history

Upgrade Path:

Tier	Cost	RAM	Storage	Notes
Basic 1GB	\$15/mo	1 GB	10 GB	May be current tier
Basic 2GB	\$30/mo	2 GB	25 GB	Recommended for 500 DAU
Basic 4GB	\$60/mo	4 GB	38 GB	If history tables grow large

Estimate for 500 DAU: \$15–30/mo

1.3 DigitalOcean Spaces — Object Storage (Avatars)

Detail	Value
Bucket	infohub
Used For	User avatar uploads
CDN	Included free

Tier	Cost	Storage	Bandwidth
Base	\$5/mo	250 GB	1 TB outbound
Overage	\$0.02/GB storage	\$0.01/GB transfer	—

With 500 users uploading ~100KB avatars = 50MB total. Negligible.

Estimate for 500 DAU: \$5/mo

1.4 Cloudflare Workers — Proxy

Detail	Value
URL	infohub-proxy.bullsvsbearsreal.workers.dev
Used For	Bypassing CloudFlare IP blocks on BitMEX, Gate.io, Crypto.com
Free Tier	100K requests/day

Estimate for 500 DAU: \$0/mo (free tier sufficient)

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2. Paid API Services

2.1 CoinMarketCap API

Detail	Value
Env Var	CMC_API_KEY
Used For	Top 500 coins, fear/greed index, global stats
Current Plan	Free (Basic) — 10,000 calls/mo, 333/day

Upgrade Path:

Plan	Cost	Calls/Month	Features
Free	\$0/mo	10,000	Basic market data
Hobbyist	\$29/mo	50,000	More endpoints
Startup	\$79/mo	—	Historical data
Standard	\$199/mo	Higher limits	Derivatives, exchange data
Professional	\$699/mo	Millions	Full historical, SLA

500 DAU with 10 page loads/day = ~5,000 CMC API calls/day = **150K/month**. Free tier (10K/mo) is insufficient. Startup or Standard needed.

Estimate for 500 DAU: \$79–199/mo

2.2 Resend — Transactional Email

Detail	Value
Env Var	RESEND_API_KEY
Used For	Auth emails, password reset, alert notifications

Plan	Cost	Emails/Month
Free	\$0/mo	3,000
Pro	\$20/mo	50,000

500 DAU: ~200 signups/mo + password resets + alert emails = ~1,000–3,000 emails/mo.

Estimate for 500 DAU: \$0/mo (free tier likely sufficient)

2.3 Sentry — Error Monitoring

Detail	Value
Env Var	NEXT_PUBLIC_SENTRY_DSN
Used For	Error tracking, performance monitoring

Plan	Cost	Errors/Month
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Developer (Free)	\$0/mo	5,000
Team	\$26/mo	50,000

Estimate for 500 DAU: \$0/mo (free tier sufficient)

2.4 Etherscan API

Detail	Value
Env Var	ETHERSCAN_API_KEY
Used For	ETH wallet balances in portfolio page
Free Tier	100K calls/day, 3 calls/sec

Low usage — only portfolio page wallet lookups.

Estimate for 500 DAU: \$0/mo (free tier sufficient)

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3. Free API Services (No Cost)

Service	Used For	Rate Limit	Risk
CoinGecko	Global market data, dominance	30 calls/min (demo), 10K/mo	May need paid (\$129/mo) at scale
DefiLlama	Protocol TVL, stablecoin data	Open	Low risk
Alternative.me	Fear & Greed Index	Open	Low risk
CryptoCompare	News aggregation	Rate limited	Low risk
ForexFactory	Economic calendar	Open	Low risk
Telegram Bot API	Alert delivery	Free	No risk
CoinMarketCal	Event calendar	Limited free	May need upgrade
CryptoPanic	News w/ sentiment	Limited free	Optional
Mempool.space	BTC mempool data	Open	Low risk

Exchange Public APIs — All 29 Exchanges (All Free)

All exchange data feeds are **free public APIs** with no authentication required:

CEX (Centralized): Binance, Bybit, OKX, Bitget, MEXC, Kraken, BingX, Phemex, Bitfinex, Deribit, KuCoin, HTX, WhiteBIT, CoinEx, Bitunix, Coinbase, BitMEX*, Gate.io*

DEX (Decentralized): Hyperliquid, dYdX, Drift, GMX, Aevo, Lighter, Aster DEX, gTrade, Extended (Starknet), edgeX, Variational

**CloudFlare-blocked from Vercel IPs — accessed via Cloudflare Workers proxy*

OAuth Providers (All Free)

Provider	Env Vars	Cost
Google OAuth	AUTH_GOOGLE_ID, AUTH_GOOGLE_SECRET	Free
Discord OAuth	AUTH_DISCORD_ID, AUTH_DISCORD_SECRET	Free
Twitter OAuth	AUTH_TWITTER_ID, AUTH_TWITTER_SECRET	Free

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4. Cost Summary — 500 Daily Active Users

Monthly Running Costs

Service	Current Cost	At 500 DAU	Notes
Vercel Hosting	\$0 (Hobby)	\$20–60	Pro plan recommended
DO PostgreSQL	~\$15	\$15–30	May need 2GB tier
DO Spaces	\$5	\$5	Minimal avatar storage
CoinMarketCap API	\$0 (Free)	\$79–199	Startup or Standard plan
Resend Email	\$0 (Free)	\$0	Under 3K emails/mo
Sentry	\$0 (Free)	\$0	Under 5K errors/mo
Etherscan	\$0 (Free)	\$0	Under 100K calls/day
CoinGecko	\$0 (Free)	\$0–129	May need paid if cached poorly
CF Workers Proxy	\$0 (Free)	\$0	Under 100K req/day
Telegram Bot	\$0 (Free)	\$0	Free forever
OAuth (Google/Discord/Twitter)	\$0	\$0	Free
TOTAL (Conservative)	~\$20	\$119/mo	Minimal upgrades
TOTAL (Comfortable)	~\$20	\$294/mo	Recommended tiers
TOTAL (Premium)	~\$20	\$423/mo	Full-featured, high limits

Annual Cost Projection

Scenario	Monthly	Annual
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Conservative	\$119	\$1,428
Comfortable	\$294	\$3,528
Premium	\$423	\$5,076

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5. Scaling Considerations

Bottlenecks at 500 DAU

1. **CoinMarketCap** — Free tier (10K calls/mo) will be exhausted within days. **First upgrade needed.**
2. **Vercel Hobby** — 100K function invocations/mo may be tight with cron jobs (liquidations run every minute = 43K/mo alone). **Second upgrade needed.**
3. **CoinGecko** — 10K calls/mo free cap may be hit if not aggressively cached. Server-side caching helps.

Cost Optimization Strategies

- **Aggressive caching:** Cache CMC/CoinGecko responses for 60s+ server-side to reduce API calls 10–50x
- **SWR deduplication:** Client-side SWR already deduplicates concurrent requests
- **Vercel Edge caching:** Use stale-while-revalidate headers to reduce function invocations
- **DB connection pooling:** Already using postgres.js with max 3 connections + 20s idle timeout

Scaling to 5,000 DAU

At 10x users, expect roughly:

Service	Cost
Vercel	\$60–200/mo
CMC	\$199–699/mo
DB	\$60–120/mo
Spaces	\$5/mo
CoinGecko	\$0–129/mo
Total	\$324–1,153/mo

6. Environment Variables Reference

Required for Production

Var	Service	Required
AUTH_SECRET	NextAuth JWT signing	Yes
DATABASE_URL	DigitalOcean PostgreSQL	Yes
CMC_API_KEY	CoinMarketCap	Yes
CRON_SECRET	Vercel Cron auth	Yes

ADMIN_API_KEY	Admin endpoints	Yes
PROXY_URL	CF Workers proxy	Yes

Optional (Enhance Features)

Var	Service	Feature
RESEND_API_KEY	Resend	Email delivery
TELEGRAM_BOT_TOKEN	Telegram	Bot alerts
DO_SPACES_* (5 vars)	DigitalOcean	Avatar CDN storage
AUTH_GOOGLE_*	Google	OAuth login
AUTH_DISCORD_*	Discord	OAuth login
AUTH_TWITTER_*	Twitter	OAuth login
ETHERSCAN_API_KEY	Etherscan	Wallet balances
NEXT_PUBLIC_SENTRY_DSN	Sentry	Error tracking
VAPID_* (3 vars)	Web Push	Browser notifications
CRYPTOPANIC_API_KEY	CryptoPanic	News sentiment
NEXT_PUBLIC_COINMARKETCAL_API_KEY	CoinMarketCal	Events calendar